

Experiments

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

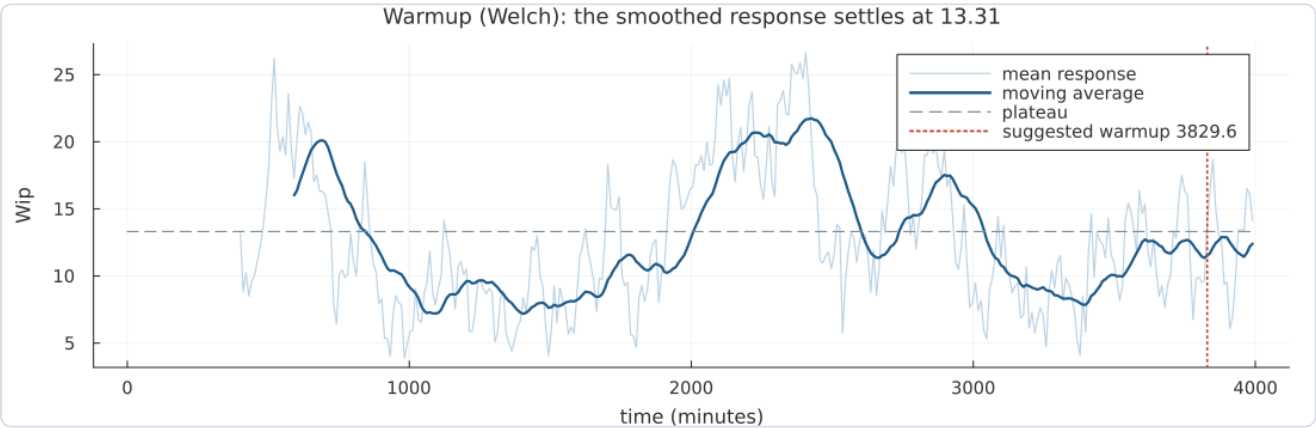
Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 | Generated: 2026-09-24 23:21:59 | Engine: DiscreteSim | Sections: 8

Experiments

Replications, a detected warmup, batch means for a single run, paired comparisons and a factorial design -- all with common random numbers.

How long the transient lasts

Welch's procedure on the wip series over 4 replications suggests discarding the first 3830 time units (plateau 13.3 ± 1.912). That is most of the horizon: for this series the diagnostic could not find a plateau -- the curve moves by 1.912 around 13.3 all the way to the end -- so the number below is a design decision, not a measurement: more replications or a longer window would settle it.



Warmup

One long run, honestly

Batch means

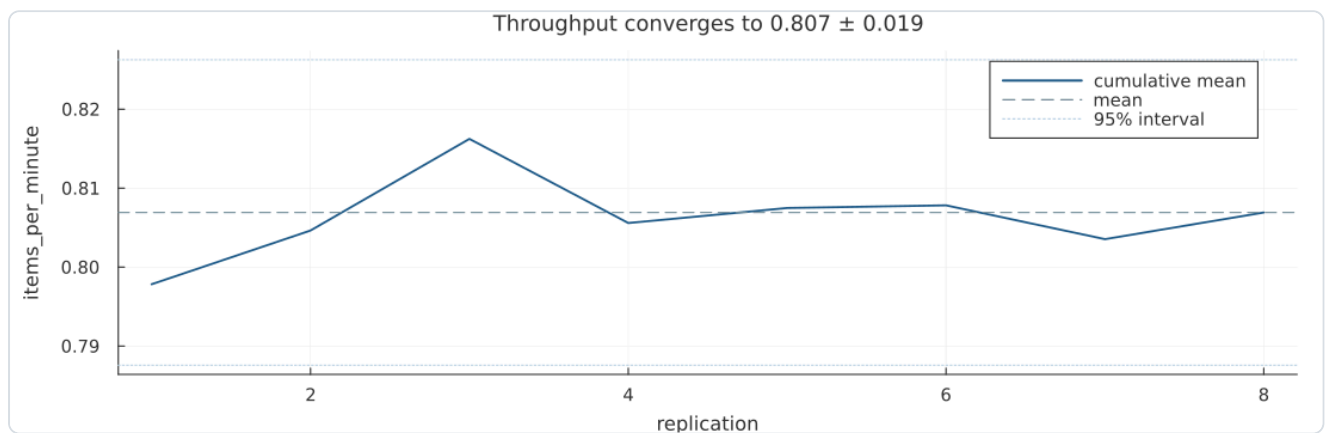
N	Mean	Half width	Lo	Hi	Batches	Lag1	Adequate
8	14.8	2.031	12.7	16.8	4	0.219	no

The replications

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
availability	0.915	0.013	ratio	1.395
wait_p95	45.5	7.774	minutes	17.1
max_queue_length	3.522	0.968	count	27.5
queue_length_mean	9.583	1.474	count	15.4
cycle_time_mean	14.8	1.575	minutes	10.7
setup	2	0	minutes	0

Metric	Mean	Half width	Unit	Rel hw pct
scrapped	141.5	13.9	count	9.851
throughput	0.807	0.019	items_per_minute	2.398
events	27050	748.3	count	2.766
span	3600	0.122	count	0.00338
reworked	427.8	17.3	count	4.053
wait_mean	14.8	1.575	minutes	10.7
completed	2905	69.7	count	2.398
wip_mean	11.9	1.522	count	12.8
utilisation	0.566	0.013	ratio	2.219
tardiness_mean	9.855	1.548	minutes	15.7



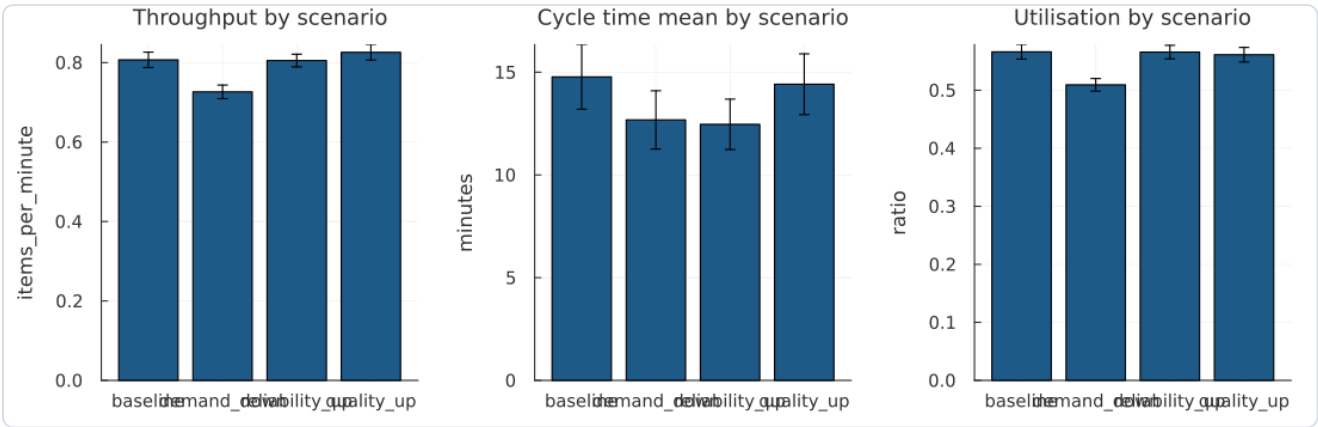
Convergence

Scenarios against the baseline

Scenarios

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	throughput	0.807	0.726	-0.081	0.00305	-9.99	worse
demand_down	cycle_time_mean	14.8	12.7	-2.092	1.367	-14.2	better
demand_down	wait_mean	14.8	12.7	-2.092	1.367	-14.2	better
demand_down	utilisation	0.566	0.509	-0.057	0.00231	-10	worse
demand_down	completed	2905	2614	-290.2	11	-9.992	worse
reliability_up	throughput	0.807	0.805	-0.0017	0.00397	-0.211	indistinguishable
reliability_up	cycle_time_mean	14.8	12.5	-2.309	1.195	-15.6	better
reliability_up	wait_mean	14.8	12.5	-2.309	1.195	-15.6	better
reliability_up	utilisation	0.566	0.566	-0.00036	0.00104	-0.064	indistinguishable
reliability_up	completed	2905	2898	-6.375	14.5	-0.219	indistinguishable
quality_up	throughput	0.807	0.826	0.019	0.00223	2.351	better
quality_up	cycle_time_mean	14.8	14.4	-0.357	0.157	-2.414	better
quality_up	wait_mean	14.8	14.4	-0.357	0.157	-2.414	better

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
quality_up	utilisation	0.566	0.561	-0.00489	0.000593	-0.863	worse
quality_up	completed	2905	2973	68.2	7.941	2.35	better



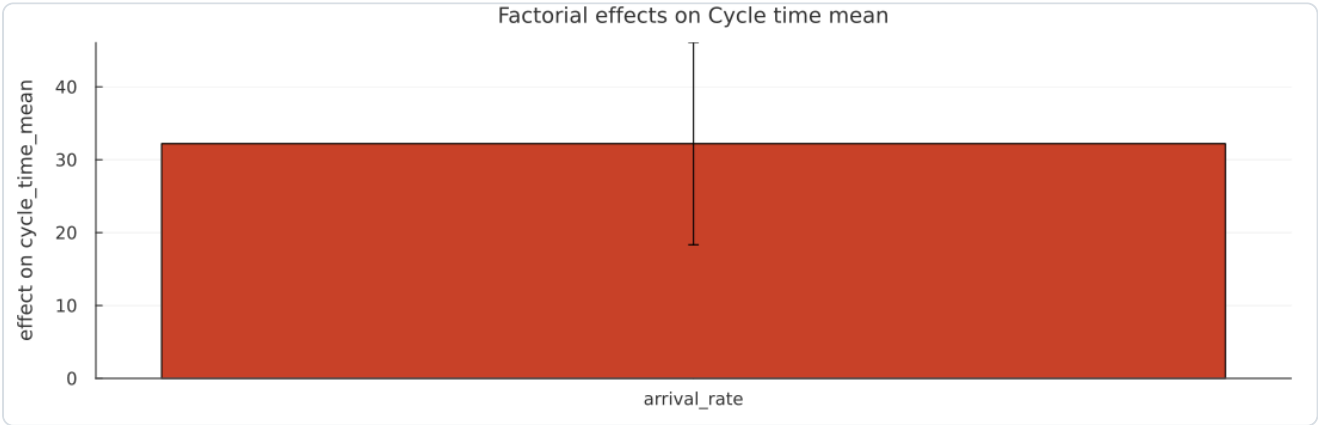
Scenarios

Best by cycle_time_mean: **reliability_up** (compared with baseline, common random numbers on every replication).

Which knob matters?

Factorial

Term	Metric	Effect	Half width	Lo	Hi	Verdict
arrival_rate	cycle_time_mean	32.2	13.9	18.3	46.1	hurts



Factorial